

DEFENSE MEDICAL COLLEGE

Final Examination in Embryology for Year I Medical Students

Time allowed: 2:00 hours.

Name: _____

Date: May 22, 2014

For each of the following **Multiple Choice Questions** choose the one **BEST** appropriate answer and mark 'X' in the appropriate box located against corresponding question number and letter in the **Separate Answer Sheet** provided.

_____ 1. Concerning **gametogenesis**, which statement is **true**?

- a) All oogonia become primary oocytes during ovulation.
- b) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
- c) Secondary oocyte completes its second meiotic division to become a mature ovum at fertilization.
- d) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.
- e) Primary spermatocytes begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.

_____ 2. Which statement is **true** about **female reproductive system**?

- a) The usual site of fertilization is the ampulla of the uterine tube.
- b) If an oocyte is fertilized, the corpus luteum involutes and degenerates.
- c) The endometrium has a wall made up of a very thick smooth muscle layer.
- d) The perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
- e) The vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.

_____ 3. With respect to the **zona pellucida**, which statement is **false**? The **zona pellucida**:

- a. Is composed of sulfated glycoproteins.
- b. Prevents polyspermy
- c. Prevents premature implantation.
- d. Functions as a porous filter
- e. Arises by delamination from the embryoblast during the 2nd week.

_____ 4. Before a **spermatozoan fertilizes an ovum**:

- a) The zona pellucida must be shed.
- b) Its glycoprotein coat must be removed from its head.
- c) Its acrosomal membrane must be impermeable.
- d) Its tail piece should be lost.
- e) It should be capacitated in the epididymidis.

_____ 5. With regard to **fertilization**, which statement is **false**? **Fertilization**:

- a) Results in variation of human species through mingling of maternal and paternal chromosomes.
- b) Causes metabolic activation of the ovum and initiates cleavage of the zygote.
- c) Restores the normal diploid number of chromosomes in the zygote.
- d) Determines chromosomal sex of the embryo.
- e) None of the above.

_____ 6. Which of the following components plays the most active role in **invading the endometrium** during **blastocyst implantation**?

- a) epiblast
- b) syncytiotrophoblast
- c) cytotrophoblast
- d) extraembryonic somatic mesoderm
- e) acrosomal reaction

_____ 7. During the **first week of development**, which statement is **true**?

- a) The prochordal plate marks the site of the future mouth.
- b) The blastocyst is superficially implanted in the uterine wall (endometrium).
- c) Developmental disturbances may cause major congenital anomalies amenable to surgery.
- d) amniotic cavity develops within the yolk sac as a space between the cytotrophoblast and the embryoblast.
- e) The embryo is a solid ball of blastomeres and is called gastrula, a process which establishes the three definitive germ layers.

_____ 8. Which of the following is/are **not** (a) derivative(s) of the **neural crest**?

- a) Schwann cells
- b) chromaffin cells
- c) melanocytes
- d) neurohypophysis
- e) parafollicular (C) cells

_____ 9. What germ layer is the **dermis** derived from?

- a) Lateral plate mesoderm
- b) Paraxial mesoderm
- c) Intermediate mesoderm
- d) Neural crest
- e) Endoderm

_____ 10. Which of the following is/are **not mesodermal in origin**?

- a) microglia
- b) bone and cartilage
- c) astrocytes
- d) cardiac muscle cells
- e) endothelium of blood and lymph vessels

_____ 11. **Intermediate mesoderm** gives rise to the

- a) neural tube
- b) heart
- c) kidneys and gonads
- d) somites
- e) notochord

_____ 12. **Sacroccygeal teratoma** is a tumor that arises from remnants of the

- a) primitive streak
- b) neural plate
- c) posterior neuropore
- d) cloacal membrane
- e) notochord

_____ 13. During the **third week of development**, which statement is **false**?

- a) somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
- b) notochord forms the primitive axis of the embryo around which the axial skeleton will form.
- c) the primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.
- d) mesoderm exists between the ectoderm and endoderm everywhere.
- e) neural tube is formed.

_____ 14. During the **fetal period**, which statement is **true**?

- a) Pupillary light reflex occurs during weeks 26-29.
- b) The amount of white fat is about 16% of body weight at 20 weeks.
- c) The spleen is the major site of erythropoiesis during the end of the 9th week.
- d) By 18 weeks the uterus is formed and canalization of the vagina is completed.
- e) The bodies of 12- week fetuses are usually completely covered with lanugo hairs.

_____ 15. With regard to the **placenta**, which statement is **true**?

- a) Deoxygenation of blood is one of the major functions of the placenta.
- b) As pregnancy advances, the placental membrane becomes progressively thicker.
- c) The fetal part of the placenta is attached to the maternal part of the placenta by the cytotrophoblastic shell.
- d) The maternal part of the placenta develops from chorionic sac.
- e) Amniochorionic membrane ruptures during labor.

_____ 16. Development of the **placental membrane** consists of all of the following **EXCEPT**:

- a) Syncytiotrophoblast b) Cytotrophoblast c) Embryonic connective tissue
d) Endothelium of the fetal vessels e) Cotyledon

_____ 17. The vessels of the umbilical cord are:

- a) 1 artery, 1 vein b) 2 arteries, 2 veins c) 1 artery, 2 veins d) 2 arteries, 1 vein e) 3 arteries, 1 vein

_____ 18. Concerning the **fetal membranes**, which statement is **true**?

- a) The yolk sac (umbilical vesicle) helps control the embryo's body temperature.
b) As the bladder enlarges, the umbilical cord involutes to form urachus and the primordial gut.
c) Primordial germ cells appear in the endodermal lining of the wall of the allantois during the 6th to 8th week.
d) Blood formation occurs in the wall of allantois during the third to fifth weeks. e) None of the above.

_____ 19. Concerning **multiple pregnancies**, which statement is **true**?

- a) Twins with one amnion, one chorion and one placenta are always monozygotic twins.
c) The frequency of monozygotic twinning shows marked racial differences.
b) The rate of dizygotic twinning increases with maternal age.
d) Dizygotic twins always have two amnions and two chorions. e) None of the above.

_____ 20. Most **birth defects** occur during:

- a) The first two weeks b) The embryonic period c) The fetal period
d) Early childhood e) The third trimester

_____ 21. With regard to **birth defects**, which statement is **false**?

- a) Clubfoot, which is due to mechanical forces in the womb, is an example of deformation.
b) Trisomy of the autosomes occurs with increasing frequency as maternal age increases.
c) Deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
d) The phenotype of Turner syndrome is female. e) Thalidomide induces limb defects.

_____ 22. Which statement is **false** with regard to **congenital malformation**?

- a) Alcohol as a teratogen is the leading cause of mental retardation.
b) Susceptibility to teratogenesis may depend on the genotype of the conceptus.
c) Heart and neural tube defects are the most likely birth defects when the mother is obese.
d) Toxoplasmosis has a characteristic feature of fetal limb hypoplasia and/or absence of extremities.
e) A vascular accident leading to bowel atresia, which is a result of morphological alteration to an already formed structure, is called disruption.

_____ 23. During the 6th through the 11th week of development a woman uses **methamphetamine**. Along with cardiovascular defects, which of the following is the most likely defect her child could be born with?

- a) Limb defects b) Anencephaly c) Cleft lip/palate
d) Masculinization of the female genitalia e) Malformation of the uterine tubes

_____ 24. Failure of **fusion of the neural arches of the developing vertebrae** could lead to:

- a) Spina bifida b) Achondroplasia c) Acromegally d) Brachycephaly e) Amelia

_____ 25. With regard to the **pharyngeal apparatus**, which statement is **false**?

- a) The third superior pouch forms the inferior parathyroid.
b) The fourth inferior pouch forms the ultimopharyngeal body.
c) The second aortic arch forms the hyoid and stapedial arteries.

d) Lesser cornu of hyoid bone develops in the third branchial arch.

e) The second pharyngeal/branchial arch is associated with muscles of facial expression.

_____ **26.** Which of the following structures develop from **endoderm of the second pharyngeal pouch**?

a) Thymus gland

b) Palatine tonsil

c) Thyroid gland

d) Superior parathyroid gland

e) Parafoallicular cells of thyroid gland

_____ **27.** With regard to the **development of the musculoskeletal system**, which statement is **true**?

a) The muscular system develops from neuroectoderm.

b) The upper limbs rotate laterally through 90° on their longitudinal axes.

c) Premature closure of the frontal (metopic) suture results in scaphocephaly.

d) Sclerotomes of the somites form the viscerocranium.

e) Xiphoid process develops during the 6th week.

_____ **28.** Which of the following **muscles** develop from the **epimeres**?

a) Intercostal muscles.

b) The strap muscles of the neck.

c) Extensor muscles of the back.

d) Rectus abdominis.

e) Quadratus lumborum.

_____ **29.** All of the following **appendages of the skin** develop from ectoderm, **except**:

a) Hairs.

b) Sweat glands.

c) Sebaceous glands.

d) Nails.

e) Arrector pili muscle.

_____ **30.** Concerning the development of the **Integumentary system**, which statement is **false**?

a) castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.

b) toenails and fingernails begin to develop at the tips of the digits postnatally.

c) teeth develop from ectoderm, mesoderm and neural crest cells.

d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.

e) the epidermis is derived from surface ectoderm.

_____ **31.** Concerning the development of the **Integumentary system**, which statement is **false**?

a) Up to the 5th week of development, the embryonic skin is only a single layer of ectoderm.

b) Leprosy results from excessive keratinization of the skin

c) Melanocytes are derived from neural crest.

d) Dermis is derived from lateral plate mesoderm.

e) None of the above.

_____ **32.** Concerning the development of the **circulatory system**, which statement is **true**?

a) *The common carotid arteries develop from truncus arteriosus.*

b) *Left umbilical vein participates in the formation of the inferior vena cava.*

c) *Right ventricle is formed by the fusion of the right and left horns of the sinus venosus.*

d) *Portal vein is a derivative of anastomosis between right umbilical and right vitelline veins.*

e) *During fetal life, the percent hemoglobin/oxygen saturation level of fetal blood the lowest in right atrium.*

_____ **33.** Which of the following is NOT associated with Tetralogy of Fallot?

a) Pulmonary stenosis

b) Pulmonary atresia

c) Ventricular septal defect

d) Right ventricular hypertrophy

e) Dextroposition of the Aorta (over riding aorta)

_____ **34.** Which of the following is NOT associated with an Atrial Septal Defect?

a) Persistent ostium secundum

b) Endocardial cushion defect

c) Sinus venosum defect

d) Probe patency

e) Transposition of the Great Vessels

_____ **35.** At birth, a child's skin appears much less pink than would be expected. The physician determines that the child's ductus arteriosus did not close. The child has a blue tint because the ductus arteriosus is shunting blood from the _____ to the _____.

- a) Right Atrium; Left Atrium b) Pulmonary Artery; Aorta c) Right Ventricle; Left Ventricle
 d) Inferior Vena Cava; Right Atrium e) Descending Aorta; Umbilical Arteries
- _____ 36. Aortic arch IV forms the _____ on the right side of the embryo and the _____ on the left.
- a) Ductus arteriosus; Right subclavian artery
 b) Right subclavian artery; Arch of aorta c) Ductus arteriosus; Pulmonary Artery
 d) Arch of aorta; Pulmonary Artery e) Arch of aorta; Ductus arteriosus

- _____ 37. During week six (lymphatic vessel development), the dilated lymph sac that forms at the lower end of the thoracic duct is called the _____.
- a) Lymphocitic Sac b) Iliac Lymph Sac c) Retroperitoneal Lymph Sac
 d) Jugular Lymph Sac e) Cisterna Chyli

- _____ 38. All of the following structures contribute to the thoracoabdominal diaphragm EXCEPT:
- a) Septum transversum b) Pleuroperitoneal fold c) Dorsal mesentery
 d) Paraxial mesoderm e) Fibrous pericardium

- _____ 39. The **dorsal mesentery** runs the entire length of the gut tube. Which of the following structures is a derivative of this mesentery and **NOT attached to the liver**?
- a) Greater omentum b) Falciform ligament c) Cloaca d) Lesser omentum e) Spleen

- _____ 40. An organ is covered only on one surface with **peritoneum and is immobile**. How would you classify this organ?
- a) Peritoneal b) Intraperitoneal c) Subperitoneal d) Retroperitoneal e) Supraperitoneal

- _____ 41. Which of the following cells of the lung develop from mesoderm?
- a) Type I Pneumocytes. b) Alveolar dust cells. c) Type II Pneumocytes.
 d) Clara cells e) Goblet cells

- _____ 42. Secretion of surfactant, which is an important substance for survival of premature infants, appears during which of the following periods of lung development?
- a) Canalicular. b) Terminal sac period c) Bronchiolar. d) Pseudoglandular e) Alveolar

- _____ 43. During the **development of the digestive system**, which statement is **true**?
- a) Parotid glands develop from ectoderm.
 b) the superior two-thirds of the adult anal canal are derived from proctodeum.
 c) the midgut may give rise to the descending colon and sigmoid colon.
 d) the dorsal pancreatic bud forms the uncinate process and part of the head of the pancreas.
 e) The anterior 2/3rd of the tongue develops at the 6th pharyngeal arch level.

- _____ 44. With regard to the **development of the digestive system**, which statement is **false**?
- a) the foregut may give rise to the lower respiratory system.
 b) the hindgut derivatives are supplied by inferior mesenteric artery.
 c) the midgut may give rise to the descending colon and sigmoid colon.
 d) The middle of the duodenum would receive blood from celiac and superior mesenteric artery.
e) during the rotation of the stomach, the lesser curvature moves to the left and the greater curvature moves to the right.

- _____ 45. The **midgut** begins with _____ and ends with _____.
- a) 1/2 jejunum; 2/3 rectum b) 1/2 jejunum; 2/3 transverse colon c) 1/2 duodenum; 2/3 transverse colon

d) 1/2 duodenum; 1/3 sigmoid colon

e) 1/2 ileum; 2/3 sigmoid colon

_____ **46.** Concerning the **development of the GIT and associated structures**, which statement is **true**?

a) Liver develops by fusion of ventral and dorsal hepatic buds.

b) Stomach rotates 270° counterclockwise, similar to the midgut.

c) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.

d) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion

develops from the local mesoderm. e) The loop of the midgut has the superior mesenteric artery in its axis.

_____ **47.** A three-week old male infant comes into the Emergency Room for **weight loss and acute dehydration**. The infant has been **projectile vomiting** and the condition is worsening. Which of the following is most likely malformation?

a) Gastric atresia

b) Pyloric stenosis

c) Annular pancreas

d) Malrotation of the gut

e) Intestinal duplication

_____ **48.** *The crura of the diaphragm develops from:*

a) Septum transversum.

b) Dorsal mesentery of the esophagus.

c) Pleuroperitoneal membrane.

d) Myotomes of the body wall.

e) None of the above.

_____ **49.** Concerning the **development of the urinary system**, which statement is **false**?

a) Urinary system begins to develop before the genital system.

b) The metanephric diverticulum is the primordium of the ureter.

c) Nephrons are derived from the metanephric mass of intermediate mesoderm.

d) Differentiation of the nephrons depends upon induction by the collecting tubules.

e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **50.** The **podocytes of Bowman's capsule** are derived from:

a) ectoderm

b) mesoderm

c) endoderm

d) endoderm and mesoderm

e) neural crest cells

_____ **51.** Which of the following is not the **adult derivative of the ureteric bud**?

a) renal pelvis

b) major calyx

c) minor calyx

d) collecting duct

e) connecting tubule

_____ **52.** Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

a) The female urethra is derived from endoderm of the urogenital sinus.

b) The urinary bladder becomes a pelvic organ during the 9th week of development.

c) In adults in both sexes, the urachus is represented by the median umbilical ligament.

d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.

e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ **53.** With regard to the **congenital anomalies of the kidneys and ureters** which statement is **false**?

a) Accessory renal arteries are end arteries.

b) Bilateral renal agenesis is associated with oligohydramnios.

c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.

d) Unilateral renal agenesis is more common in males than females.

e) None of the above.

_____ **54.** During the **development of the kidney systems**, which statement is **true**?

a) The transitional epithelium lining the urethra is derived from neuroectoderm.

b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.

c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.

- d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated appearance.

_____ **55.** With regard to the development of the **endocrine system**, which of the following statements is **false**?

- a) The pituitary gland is purely ectodermal in origin.
b) The thyroid gland is the first endocrine gland to develop in the embryo.
c) The pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.
d) C cells of the thyroid gland differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
e) The oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.

_____ **56.** With regard to the **development of gonads**, which statement is **false**?

- a) Sex of an embryo is determined genetically at the time of fertilization.
b) Germ cells do not normally appear in the genital ridges until the 6th week of development.
c) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.
d) The surface epithelium of the female gonad develop into oogonia. e) None of the above.

_____ **57.** During the development of the **female reproductive system**, which statement is **false**?

- a) The urogenital folds remain unfused and form the labia minora.
b) The cranial portion of the paramesonephric ducts form the uterine tubes.
c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
d) The phallus is formed from the mons pubis. e) No oogonia are formed postnatally.

_____ **58.** A structure found within the adult ***female pelvis formed from the gubernaculum*** is

- a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

_____ **59.** During the development of the **male reproductive system**, which statement is **false**?

- a) Hypospadias is constantly associated with exstrophy of urinary bladder.
b) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
c) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
d) Sertoli cells, derived from mesoderm, constitute most of the seminiferous epithelium in the fetal testis.
e) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from the mesonephric ducts.

_____ **60.** Concerning the **development of the genital system**, one statement is **false**?

- a) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
b) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora
c) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as chordee.
d) Bilateral cryptorchidism may result in sterility. e) None of the above.

_____ **61.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The prostatic utricle is homologous to the vagina.
b) The cranial end of mesonephric duct in males may persist as paradidymis.
c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **62.** Which of the following inhibits the growth of the **paramesonephric duct (males)**?

a) Testosterone b) Leydig Cells c) Dihydrotestosterone d) Estrogens e) Müllerian substances (MIFs)

_____ **63.** Concerning the **development of the genital system**, one statement is **true?**

- a) The labia minora arise from urogenital folds.
- b) In the female, failure of urethral folds to fuse completely results in hydrocele.
- c) The uterine tubes of the adult female are derived from uterovaginal primordium.
- d) The most common cause of testicular feminization [Androgen Insensitivity] syndrome is inadequate production of testosterone and müllerian-inhibiting factor.
- e) None of the above.

_____ **64.** During the development of the **reproductive system**, which statement is **true?**

- a) Genital tubercle during weeks 3-6 becomes the clitoris.
- b) The lower portion of the vagina is derived from sinovaginal bulb.
- c) In the male, failure of urethral folds to fuse completely results in cryptorchidism.
- d) Dihydrotestosterone inhibits the growth of the paramesonephric ducts in males.
- e) In Turner's syndrome the vagina ends in a blind pouch and testes are lodged in the inguinal canal.

_____ **65.** A patient is born with karotype **46, XY** but has no problem with production of testosterone or müllerian inhibiting hormone. Their **vagina ends in a blind pouch** and **testes are lodged in the inguinal canal**. Which of the following is the best description of this patient:

- a) Male pseudo-hermaphrodite b) Female pseudo-hermaphrodite c) Androgen insensitivity syndrome
- d) Turner's syndrome e) Klinefelter syndrome

_____ **66.** During the **development of the testes**, which statement is **false?**

- a) Interstitial cells (of Leydg) begin to secrete androgenic hormones by the end of the third trimester.
- b) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.
- c) The fetal testis produces antimüllerian hormone by sertoli cells at 6-7 weeks
- d) Sertoli cells are derived from the surface epithelium of the testis.
- e) None of the above.

_____ **67. Development of the testis** is induced by the sry-gene, which encodes testis-determiningfactor (TDF). TDF causes all of the following **EXCEPT**:

- a) The sex cords continue to proliferate to form the testis.
- b) Leydig cells are secreting testosterone thus have very little SER.
- c) Towards the hilum, the cords grow into a network that will give rise to the rete testes.
- d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium.
- e) Sex cords become seminiferous cords that form seminiferous tubules (Sertoli cells), straight tubules, and rete testes

_____ **68.** During the **development of the central nervous system** zone of the wall of the neural tube to be occupied by neuroblasts forming the gray matter is:

- a) Marginal zone b) Ventricular zone c) Mantle zone d) Ependymal zone e) Neuroepithelial zone

_____ **69.** The **epoophoron** and **paroophoron**, which are important as cysts, arise from:

- a) The Mullarian ducts b) The paramesonephric ducts c) The mesonephric ducts
- d) Leydig hormones e) Wolffian hormones

_____ **70.** The **lower portion of the vagina** is derived from:

- a) Sinovaginal bulb b) Uteral lumen c) Hymen d) Urogenital sinus e) Vaginal plate

_____ **71.** The **urorectal septum** divides the _____ from the _____.

- a) Anorectal canal; Urogenital sinus b) Urogenital sinus; Mesonephric duct c) Urachus; Allantois

d) Urinary bladder; Allantois e) Mesonephric duct; Ureteric bud

_____ **72. Primary brain vesicle** that remains relatively unchanged as compared to the others is:

- a) Rhombencephalon b) Prosencephalon c) Telencephalon
- d) Mesencephalon e) Diencephalon

_____ **73.** Concerning the **development of the genital system**, one statement is **false**?

- a) Gubernaculum does not play any role during the descent of ovaries.
- d) The persistent processus vaginalis in the female is called canal of Nuck.
- c) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora.
- b) Inguinal canals develop in both sexes. e) None of the above.

_____ **74.** The **neuroepithelial cells of the neural tube** gives rise to all of the following cells of the brain except:

- a) Microglia. b) Astrocytes. c) Oligodendrocytes.
- d) Purkinje cells of cerebellar cortex. e) Ependymal cells.

_____ **75.** With regard to **development of the eye**, which statement is **true**?

- a. The hyaloid canal is found in the vitreous body.
- b. Aqueous humor is produced by the choroid plexus.
- c. Failure of choroid fissure to close results in congenital aphakia.
- d. Sclera is continuous with the dura of the central nervous system.
- e. Abnormal development of canal of Schlemm or the iridocorneal filtration angle causes congenital cataract.

_____ **76.** Which of the following structures in the **eye** is derived from **surface ectoderm**?

- a) retina b) dilator and sphincter pupillae muscles of iris
- c) corneal endothelium d) vitreous body e) lens
- e. Aqueous humor is produced by the choroid plexus.

_____ **77.** Abnormal development of **canal of Schlemm or the iridocorneal filtration angle** causes

- a. coloboma iridis b. congenital cataract c. congenital glaucoma
- d. cyclopia e. microphthalmia

_____ **78.** With regard to **development of the ear**, which statement is **true**?

- a. The middle ear cavity develops from pharyngeal pouch 1.
- b. Perilymph enters the subarachnoid space via the ductus reuniens.
- c. Otic placode and neural crest form the vestibulocochlear ganglion.
- d. The saccular portion of the otic vesicle gives rise to the endolymphatic duct.
- e. Utricular portion of the otic vesicle, which is endodermal in origin, gives rise to the organ of Corti.

_____ **79.** With regard to **development of the ear**, which statement is **true**?

- a. The spiral organ of Corti is derived from neural crest.
- b. The middle ear cavity develops from the otic vesicle.
- c. The utricular portion of the otic vesicle gives rise to cochlear duct.
- d. Perilymph enters the subarachnoid space via the endolymphatic sac. e. None of the above

_____ **80.** The **middle ear cavity**

- a) is of mesodermal origin b) develops from the otic vesicle c) develops from pharyngeal arch 1
- d) develops from pharyngeal pouch 2 e) develops from pharyngeal pouch 1